



x

1

7

x

10

0

1

0

1







PARADOXES

ESPRESSO

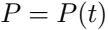




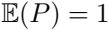




BEAR







100%







2023-10-21

$$= \underbrace{\frac{p_t^2}{2 \log(2)} \pi \lambda \left(\frac{h \varphi_{\text{RX}}}{\sin^2(\epsilon)} \right)^2}_{k \triangleq} \exp \left\{ - \tau^2 \underbrace{v_{\text{sat}}^2 \sin^4(\epsilon) / (h^2 \varphi_{\text{RX}}^2)}_{D \triangleq} \log(2) / 2 \right\},$$































Fig 1(a): True $P(t)$, GPR estimate & forecast, and 100-sample MA & forecast

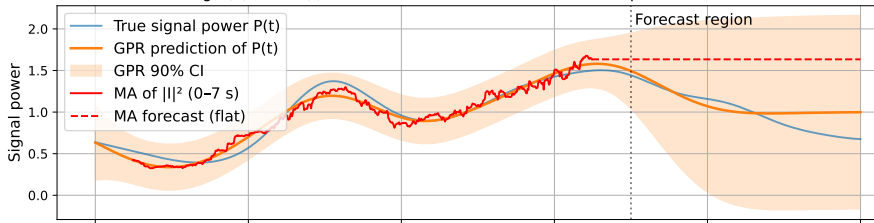
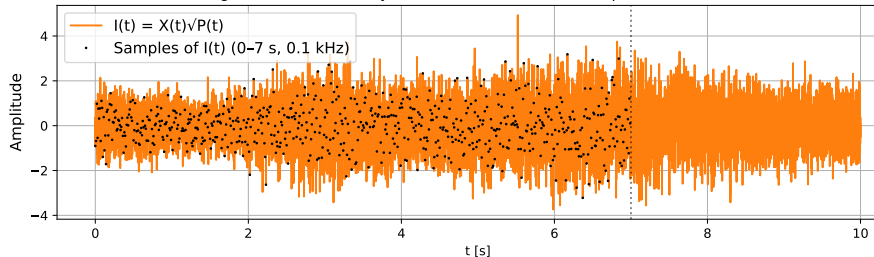


Fig 1(b): Non-stationary Gaussian noise $I(t)$ with sampled observations











expanding world + new world







Q1 2019









$$S_{\text{RX}}(t) = \frac{A}{l(d_0)} \sqrt{\frac{K}{K+1}} e^{-2i\pi\tau_0(t)f_c} e^{-2i\pi\theta} + \sum_{j=1}^{100} \frac{A}{l(d_j) \sqrt{100} \sqrt{K+1}} e^{-2i\pi\tau_j(t)f_c} e^{-2i\pi\theta},$$

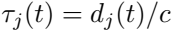


















QWERTYUIOP

$$v = \sqrt{\frac{GM}{h + R_{\oplus}}},$$

WORLDWIDE







20

—

12

.

100



$$T_c = \frac{1}{8D_g} \approx 10^{-3}$$







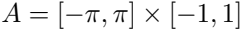




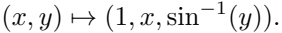


















$$1 = e^{-\frac{\pi^2}{2\sqrt{P}}}$$





$$] = 1 - \left(1 - e^{-\frac{\pi^2}{2\sqrt{P/N}}} \right)^N,$$





192







$$t \rightarrow \text{order} \rightarrow \frac{\sin(2\pi B t)}{\pi t},$$





$$\mathbb{Z} \ni i \mapsto \frac{\sin\left(2\pi \frac{B_L}{f_c} i\right)}{\pi i},$$

$$i \mapsto \delta[i] - \frac{\sin(2\pi \frac{B_H}{f_c} i)}{\pi i},$$



গণপ্রজাতন্ত্রী বাংলাদেশ

WAVELENGTH



WINNERS

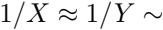
$$\mu = \exp\left(\mu_{\text{LN}} + \frac{\sigma_{\text{LN}}^2}{2}\right)$$

$\sin^2 \theta = 1 - \sin^2 \theta$









WORLD OF



1997-1998

A 16x16 grayscale pixelated image of a stylized letter 'Z'. The letter is formed by a central vertical column of black pixels, with horizontal bars at the top and bottom. The top bar is composed of black and dark gray pixels, while the bottom bar is composed of dark gray and black pixels. The vertical stem is primarily black, with some gray pixels on the left side. The overall shape is a bold, blocky 'Z' centered within the 16x16 grid.

A 16x16 grayscale pixelated image of the number 7. The image is composed of a grid of pixels with varying intensities of gray, ranging from white to black. The number 7 is formed by a thick vertical stroke on the right side and a horizontal stroke at the top. The pixels are arranged in a way that creates a blocky, digital appearance. The background is white, and the number itself is rendered in shades of gray, with the darkest pixels forming the core of the number.

Age Group	Percentage
18-24	~15%
25-34	~45%
35-44	~35%
45-54	~30%
55-64	~25%
65-74	~20%
75-84	~15%
85+	~5%

$$S(t) = \sum_{n=0}^{N-1} x[n]$$













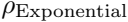


$$H = \begin{pmatrix} -a & -0 & -1 & 0 & 0 & 0 & ar & 0 & r \\ 0 & 0 & 0 & -a & -0 & -1 & 0 & 0 & 0 \\ -c & -b^2/a & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -c & -b^2/a & -1 & cr & rb^2/a & r \\ a & 0 & -1 & 0 & 0 & 0 & ar & 0 & -r \\ 0 & 0 & 0 & a & 0 & -1 & 0 & 0 & 0 \\ -c & b^2/a & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -c & b^2/a & -1 & -rc & rb^2/ar & -r \end{pmatrix}$$

$$\lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X \geq x)}{x},$$







$$= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = \lim_{x \rightarrow \infty} - \frac{\log(e^{-x/\theta})}{x}$$

$$= \lim_{x \rightarrow \infty} \frac{-x/\theta}{x} = 1/\theta.$$





$$= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = \lim_{x \rightarrow \infty} - \frac{\log(1 - \gamma(k, x/\theta)/\Gamma(k))}{x}$$

$$= \lim_{x \rightarrow \infty} - \frac{\log(\Gamma(k, x/\theta)/\Gamma(k))}{x}$$

$$\begin{aligned}
 &\stackrel{(a)}{=} \lim_{x \rightarrow \infty} - \frac{\log(\Gamma(k, x/\theta))}{x} = \lim_{x \rightarrow \infty} - \frac{\log(x^{k-1} e^{-x/\theta})}{x}
 \end{aligned}$$

$$= \lim_{x \rightarrow \infty} - \frac{\log(e^{-x/\theta})}{x} = 1/\theta.$$



$$\lim_{x \rightarrow \infty} \frac{\Gamma(s, x)}{x^{s-1} e^{-x}} = 1.$$







$$\begin{aligned}
&= \lim_{x \rightarrow \infty} - \frac{\log \mathbb{P}(X > x)}{x} = - \lim_{x \rightarrow \infty} \frac{\log \operatorname{erfc} \left(\frac{x - \mu}{\sigma \sqrt{2}} \right)}{x} \stackrel{(b)}{\geq} - \lim_{x \rightarrow \infty} \frac{\log \left(\sqrt{\frac{e}{2\pi}} e^{-2 \left(\frac{x - \mu}{\sigma \sqrt{2}} \right)^2} \right)}{x}
\end{aligned}$$

$$\geq -\lim_{x \rightarrow \infty} \frac{\log \left(e^{-2 \left(\frac{x-\mu}{\sigma \sqrt{2}} \right)^2} \right)}{x} = -\lim_{x \rightarrow \infty} \frac{-2 \left(\frac{x-\mu}{\sigma \sqrt{2}} \right)^2}{x} = \frac{1}{\sigma^2} \lim_{x \rightarrow \infty} \frac{x^2 - 2x\mu + \mu^2}{x} = \infty,$$



$$\psi(x) \geq \sqrt{\frac{2e}{\pi}} \frac{\sqrt{\beta-1}}{\beta} e^{-\beta x^2},$$















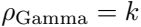




















WELCOME TO THE WORLD OF



Q1 2024

2023年12月



$$\mathbb{E}[(S_1 + S_2)^2] = \int_0^1 (\cos(2\pi t) + \cos(2\pi t + \pi))^2 dt = \int_0^1 0 dt = 0,$$



1991 + 1991

$$= \int_0^1 \cos^2(2\pi t) dt + \int_0^1 \cos^2(2\pi t + \pi) dt$$

$$= \int_0^1 2 \cos^2(2\pi t) dt = \int_0^1 \cos(4\pi t) dt + 1 = 1.$$





$$E[S_1^2 + S_2^2] = E[S_1^2] + E[S_2^2] = E[S_1^2] + E[S_2^2]$$

1991 + 1991 + 1991

1992-93

$$\mathbb{E}[S_1 S_2] = \int_0^1 \cos(2\pi t) \cos(2\pi t + \pi) dt = \int_0^1 -\cos^2(2\pi t) dt = -\frac{1}{2} \int_0^1 \cos(4\pi t) dt - \frac{1}{2} = -\frac{1}{2},$$

01

,

02

E

1

0

,

2

1

1

2024 + 2025

2023-04-20

$$\mathbb{E} \phi_1, \phi_2 [\mathbb{E} [S_1^2 + S_2^2]] = \mathbb{E} \phi_1 [\mathbb{E} [S_1^2]] + \mathbb{E} \phi_2 [\mathbb{E} [S_2^2]] + \mathbb{E} \phi_1, \phi_2 [\mathbb{E} [S_1 S_2]]$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_1))^2 dt d\phi_1 + \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_2))^2 dt d\phi_2$$

$$+ \frac{1}{2\pi} \int_0^{2\pi} \int_0^{2\pi} \int_0^1 \cos(2\pi t + \phi_1) \cos(2\pi t + \phi_2) dt d\phi_1 d\phi_2$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_1))^2 dt d\phi_1 + \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 (\cos(2\pi t + \phi_2))^2 dt d\phi_2 = 1/2 + 1/2 = 1.$$

1992



212122222





$${}_3F_2(1, 1, 1 + b; 2, 2; x) = \sum_{n=0}^{\infty} \frac{(1)_n (1)_n (1 + b)_n}{(2)_n (2)_n} \frac{x^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{(1+b)_n}{(n+1)^2 n!} x^n = \frac{1}{b!} \sum_{n=0}^{\infty} \frac{(n+1)_b}{(n+1)^2} x^n$$

$$\underline{\underline{(a)}} \quad \frac{1}{b!} \sum_{n=0}^{\infty} \frac{\sum_{k=1}^b \begin{bmatrix} b \\ k \end{bmatrix} (n+1)^k}{(n+1)^2} x^n$$

$$\begin{aligned}
 &= \frac{1}{b!} \sum_{k=1}^b \left[\begin{matrix} b \\ k \end{matrix} \right] \sum_{n=0}^{\infty} \frac{x^n}{(n+1)^{2-k}} \stackrel{(b)}{=} \frac{1}{b!} \sum_{k=1}^b \left[\begin{matrix} b \\ k \end{matrix} \right] \frac{\operatorname{Li}_{2-k}(x)}{x},
 \end{aligned}$$



